REAL ESTATE / DATA PROJECT

Real Estate Investment Advisor

Project report: data quality, target design, exploratory findings, and deployed application

Prepared 2026-11-02

Executive summary

This project demonstrates an end-to-end housing-listing workflow: clean and engineer listing data, explore its patterns, define illustrative investment targets, and serve results through Streamlit. The source data is synthetic-looking and prices are unusually similar across locations. Treat the app as a portfolio and teaching demonstration, not as Indian market research or investment advice.

250,000	20	42	500
listings	states	cities	localities

Project deliverables

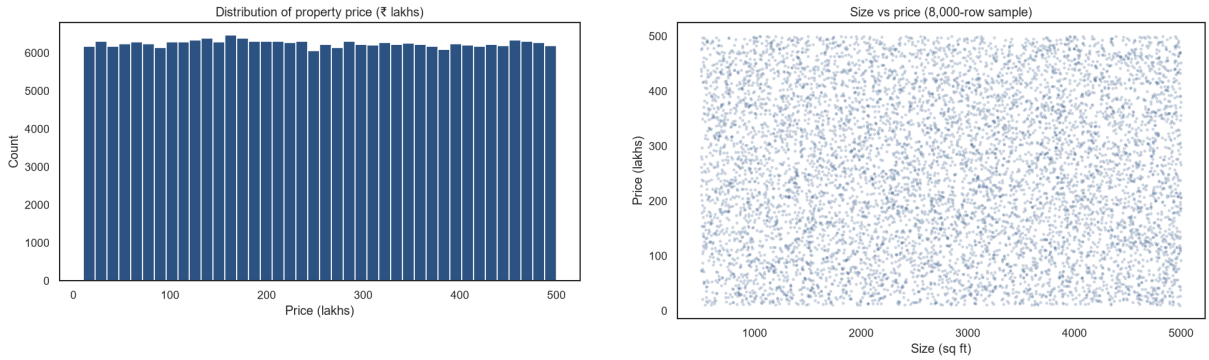
- Streamlit pages: Predict, Explore, and Insights.
- Processed dataset, EDA charts, preprocessing report, and methodology notes.
- Public source repository and hosted Streamlit application.
- Rule-based prediction fallback; trained model pipelines and benchmark metrics are not currently available.

1. Data and preparation

The source table contains 250,000 records and 23 columns, with no missing values or duplicate rows. The analysis covers 20 states, 42 cities, and 500 localities.

Check	Finding	Treatment
Price per square foot	Original field is rounded and in lakhs/sq ft.	Recomputed in INR as price in lakhs x 100,000 / area.
Floor values	46.52% have floor number above total floors.	Kept; flagged and capped only for floor ratio.
INR/sq ft outliers	19,723 records, or 7.89%, flagged by IQR.	Kept in the data; flags retained for analysis.
Property age	Matches 2025 minus year built.	Recomputed age and mismatch flag stored.

An infrastructure proxy combines nearby schools, nearby hospitals, and ordinal public-transport access. It is not an official infrastructure index. No crime score or listing date is present in the source.



Price distribution and size-price relationship in the source table.

2. Targets and modelling

The project defines targets from explicit rules so the demo is reproducible. They are not independent evidence of real-world investment performance.

Investment score point	Rule
1	INR/sq ft at or below the city median.
2	At least 3 bedrooms (BHK).
3	Ready to move.
4	Infrastructure proxy above the dataset median.
5	Amenity count above the dataset median.
6	High public-transport access.

Good Investment is labelled positive at score >= 3. The dataset has 146,530 positive and 103,470 negative records. The label is rule-derived, so high classification scores would not mean that a model discovered an independently validated investment signal.

Illustrative five-year value

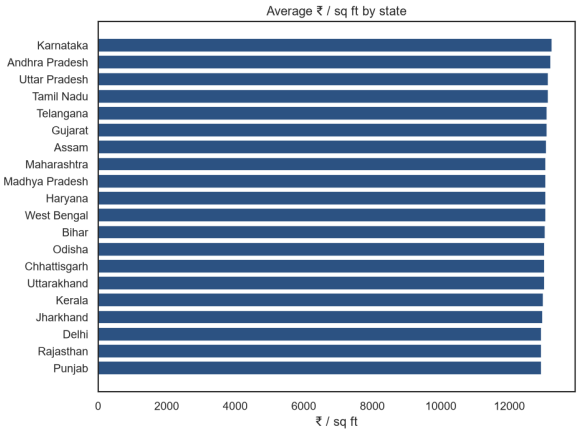
Future price = current price x (1 + annual rate)^5. The annual rate combines a city-tier base (6%, 8%, or 10%), a property-type adjustment (0 to 1.5 percentage points), and up to 2 percentage points from the infrastructure proxy; the final rate is clipped to 5%-13%. The mean target rate in the processed dataset is 9.78%.

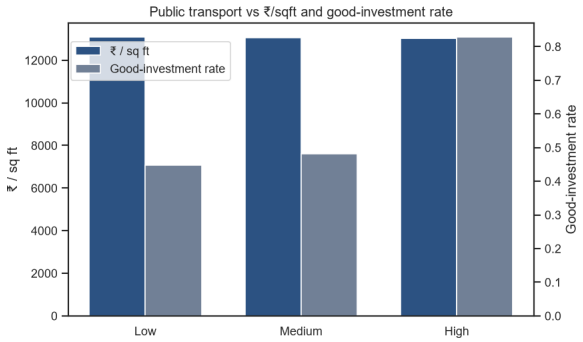
Model status: the published app currently uses the documented rule-based fallback because trained classifier/regressor pipelines and a completed model-comparison report are absent. Do not interpret fallback outputs as trained-model predictions.

3. Exploratory findings and use

The strongest conclusion is a limitation of the source data: prices and price per square foot barely vary by city or state. City mean prices range only about 250.8-258.5 lakhs, and state mean price per square foot ranges about INR 12,931-13,252. Amenities, parking, furnishing, facing, nearby schools, and hospitals show little price separation.

The good-investment rate rises with high public transport partly by construction, because transport is one of the six label rules. This is a target-design relationship, not causal evidence that transit access will raise returns in a real market.





State-level price similarity and transport-related target rate.

Application and intended use

Predict screens a hypothetical listing and estimates value under the project rate table. Explore filters and compares listings. Insights presents EDA charts and the data caveats. Use the app for portfolio demonstration and data-workflow education only; it is not a valuation, purchase recommendation, or substitute for local comparable sales, title checks, and site inspection.

Live Streamlit app	Open deployed app
Public GitHub repository	View source code

Sources: project preprocessing report, methodology, findings, and EDA charts. Figures describe the supplied synthetic dataset only.